



CVR COLLEGE OF ENGINEERING
(An UGC Autonomous Institute, Accredited by NAAC with 'A' Grade)

Department of CSE
B. Tech CSE-E III Year II Semester
Industrial Oriented Mini Project

Project Title:

**Time Series Forecasting and Modeling of Food Demand Supply Chain Using
Regressor Analysis**

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Abstract

Accurate food demand forecasting is crucial for meal delivery services because food is perishable and small prediction errors can cause losses. Existing platforms like Blue Yonder, SAP Integrated Business Planning, Oracle Demantra, and Infor Demand Management mainly use traditional statistical methods. These often fail to capture complex demand patterns, leading to overstocking or shortages. This study proposes a time series–based forecasting approach using machine learning and deep learning models. The dataset from Genpact includes weekly orders, pricing, promotions, and meal categories. Lag features and moving averages are used to model temporal patterns effectively. Random Forest, CatBoost, and LightGBM regression models were evaluated for food demand forecasting. Based on validation metrics, LightGBM demonstrated superior accuracy and stability compared to the other models. This improved predictive performance contributes to better inventory planning and supply chain optimization.

Introduction

- Food demand forecasting is essential for efficient supply chain management in meal delivery services. Since food is perishable, inaccurate predictions can cause wastage, losses, and customer dissatisfaction. Traditional statistical models often fail to capture complex demand patterns.
- FoodDemandAI is a machine learning–based forecasting system designed to predict weekly meal demand accurately. It uses historical order data, pricing, promotional details, and fulfillment center information to analyze trends and patterns. Advanced feature engineering techniques such as lag features and moving averages are used to model temporal dependencies.
- The system compares models such as Random Forest, CatBoost, and LightGBM. Results show that LightGBM provides higher accuracy and stability, helping reduce wastage and improve supply chain decisions.

Literature Review:

S.No	Existing System	Description	Key Features	Limitations
1	Blue Yonder	AI-based supply chain and demand forecasting platform	Demand planning, inventory optimization, AI-driven analytics	Expensive, complex implementation, limited customization for small businesses
2	SAP IBP	Cloud-based integrated business planning solution	Real-time forecasting, scenario planning, supply chain integration	High cost, requires technical expertise, complex setup
3	Oracle Demantra	Demand management and advanced forecasting tool	Statistical forecasting models, demand sensing, automated planning	Relies heavily on traditional models, less focus on deep learning
4	Infor Demand Management	Predictive analytics platform for supply chain planning	Machine learning forecasting, demand collaboration tools	Limited flexibility, subscription cost high
5	Traditional Statistical Models (ARIMA, MA)	Time series forecasting using historical data patterns	Simple implementation, interpretable results	Cannot capture complex nonlinear patterns, lower accuracy

Objectives

- Accurate Demand Prediction
- Reduction of Food Wastage
- Prevention of Stockouts
- Inventory Optimization
- Cost Reduction & Profit Maximization
- Data-Driven Decision Making
- Seasonality & Trend Analysis
- Scalable and Automated Forecasting System
- Continuous Model Improvement Using Machine Learning

Existing Systems

Blue Yonder:

- AI-driven supply chain and demand forecasting platform used by large enterprises.
- Provides advanced analytics, inventory optimization, and automated planning.
- Very expensive and complex to implement for small and medium businesses.
- Requires high infrastructure and skilled professionals for maintenance.

SAP IBP:

- Cloud-based integrated business planning solution for supply chains.
- Supports real-time forecasting and scenario-based planning.
- High implementation cost and complex configuration process.
- Requires technical expertise and is less flexible for customized forecasting needs.

Problem Statement

- The current food demand planning process in many meal delivery services relies on traditional statistical models or manual estimation techniques. While these methods provide basic forecasts, they often fail to capture complex temporal patterns, seasonal variations, and promotional impacts. This leads to overstocking, food wastage, stock shortages, and financial losses due to inaccurate demand predictions.
- Existing enterprise solutions are expensive and complex, making them unsuitable for small and medium businesses. They often lack advanced deep learning capabilities to capture complex demand patterns. Therefore, a scalable forecasting system using machine learning and deep learning is needed for accurate and reliable demand prediction.
- **Problem Domain Knowledge :**
Food Demand Forecasting falls under the domains of Time Series Analysis, Machine Learning, and Supply Chain Analytics. It involves analyzing historical order data, pricing, promotions, and categorical features to predict future demand. Accurate forecasting requires handling temporal dependencies, seasonality, and external influencing factors to ensure optimal inventory management and reduced food wastage.

Functional And Non-Functional Requirements:

Functional Requirements

1. Data loading & preprocessing
2. Feature engineering (lag features, moving averages)
3. Model training (Random Forest, CatBoost, LightGBM)
4. Model evaluation (MAE, RMSE)
5. Weekly demand forecasting

Non-Functional Requirements

1. Prediction accuracy
2. High performance
3. Scalability
4. Fast response time
5. Maintainability

Hardware & Software Requirements

Hardware Requirements:

- **Processor:** Intel i5
- **RAM:** 8 GB
- **Storage:** 256 GB SSD
- **System Type:** 64-bit system
- Internet connection

Software Requirements:

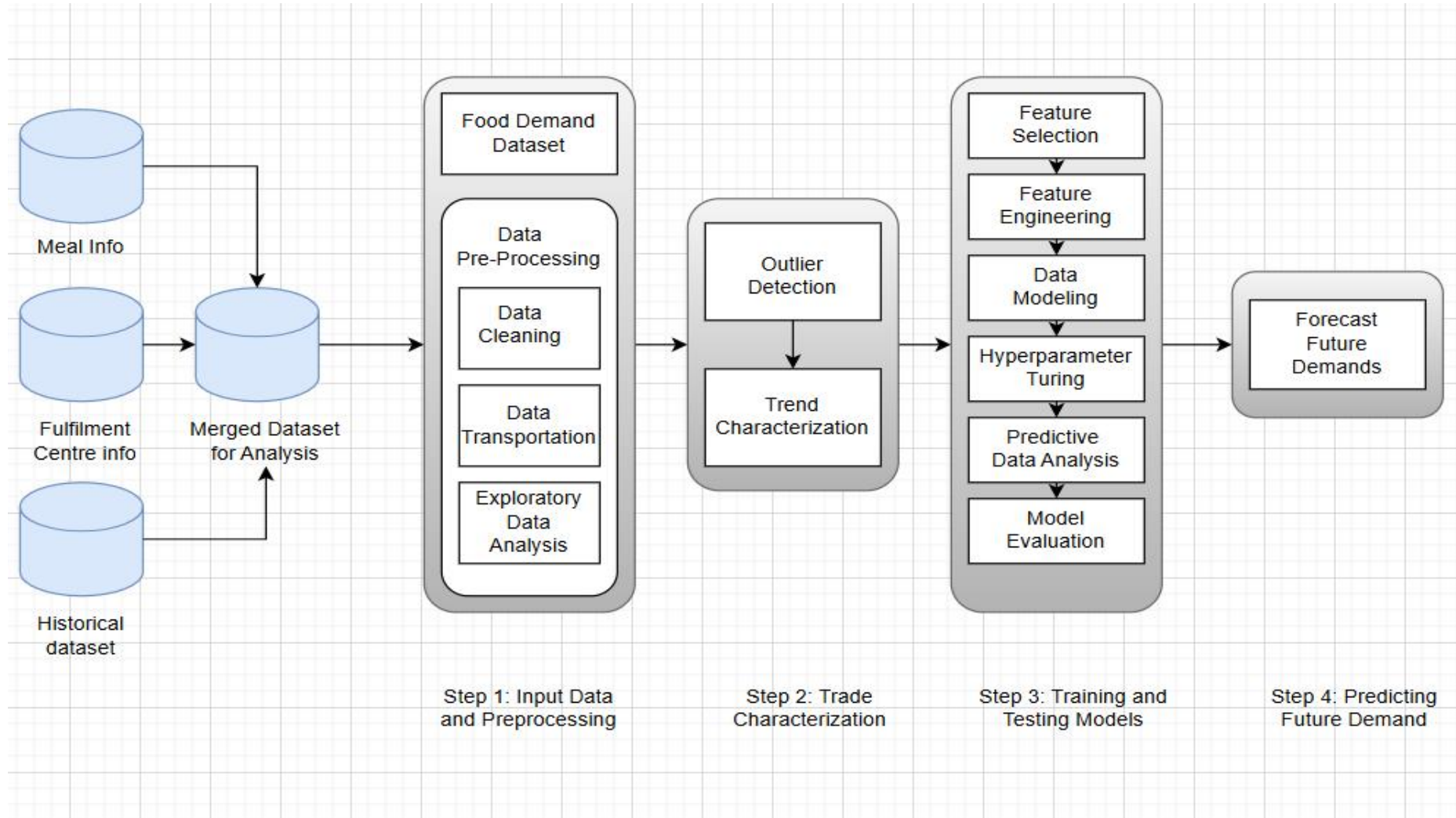
- **Operating System:** Windows
- **Programming Language:** Python 3.x
- **IDE:** Visual Studio Code (VS Code)
- **Machine Learning Framework:** LightGBM
- **Libraries:** NumPy, Pandas, Matplotlib, Scikit-learn
- **Version Control:** Git

Proposed Methodology

The proposed system uses a time series–based machine learning and deep learning approach to predict weekly food demand.

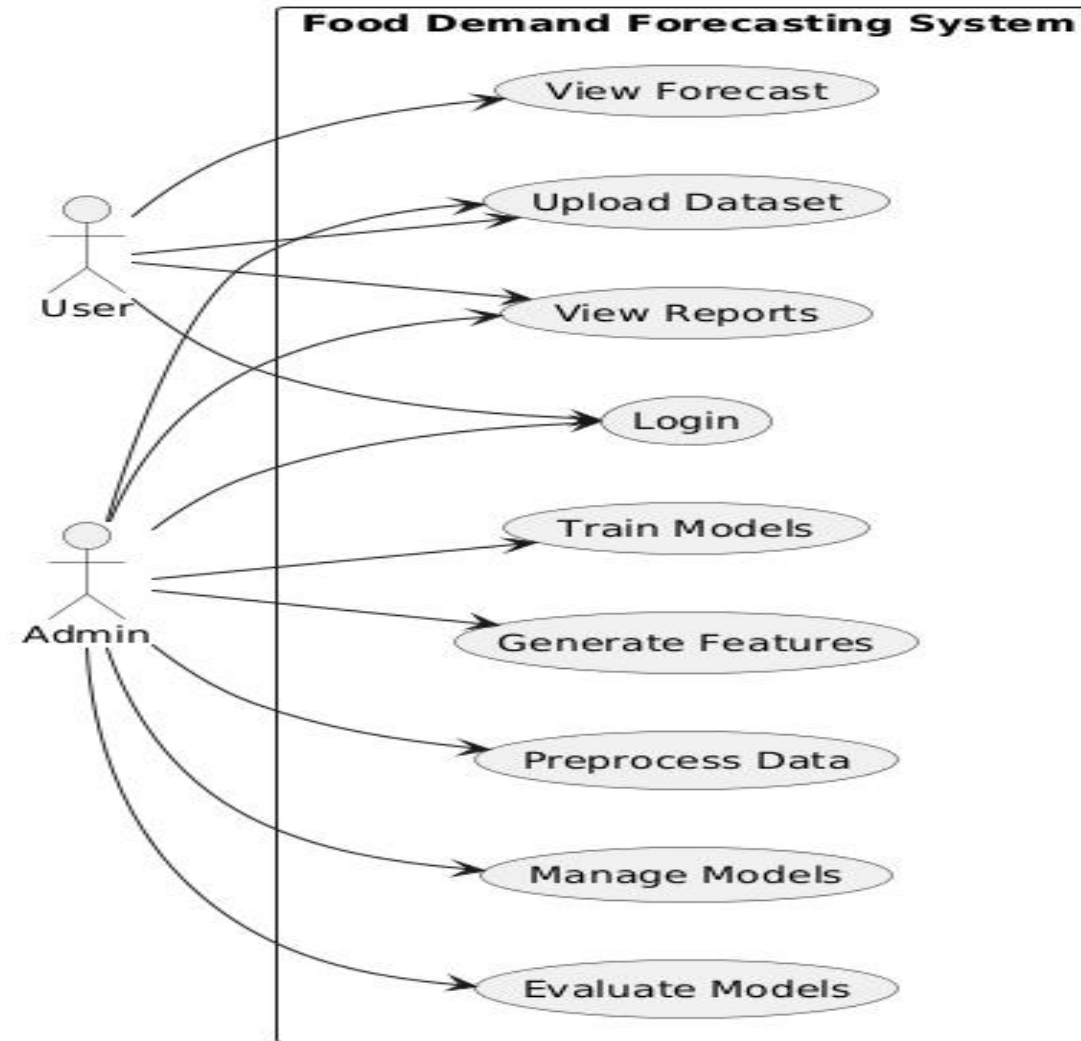
- **Data Collection:** Historical order, pricing, and promotion data are used.
- **Data Preprocessing:** Missing values are handled and categorical features are encoded.
- **Feature Engineering:** Lag features and moving averages are created to capture trends.
- **Model Building:** Random Forest, CatBoost, and LightGBM models are trained.
- **Model Evaluation:** Models are compared using MAE and RMSE.
- **Forecasting:** The best model generates demand predictions for better inventory planning.

System Architecture



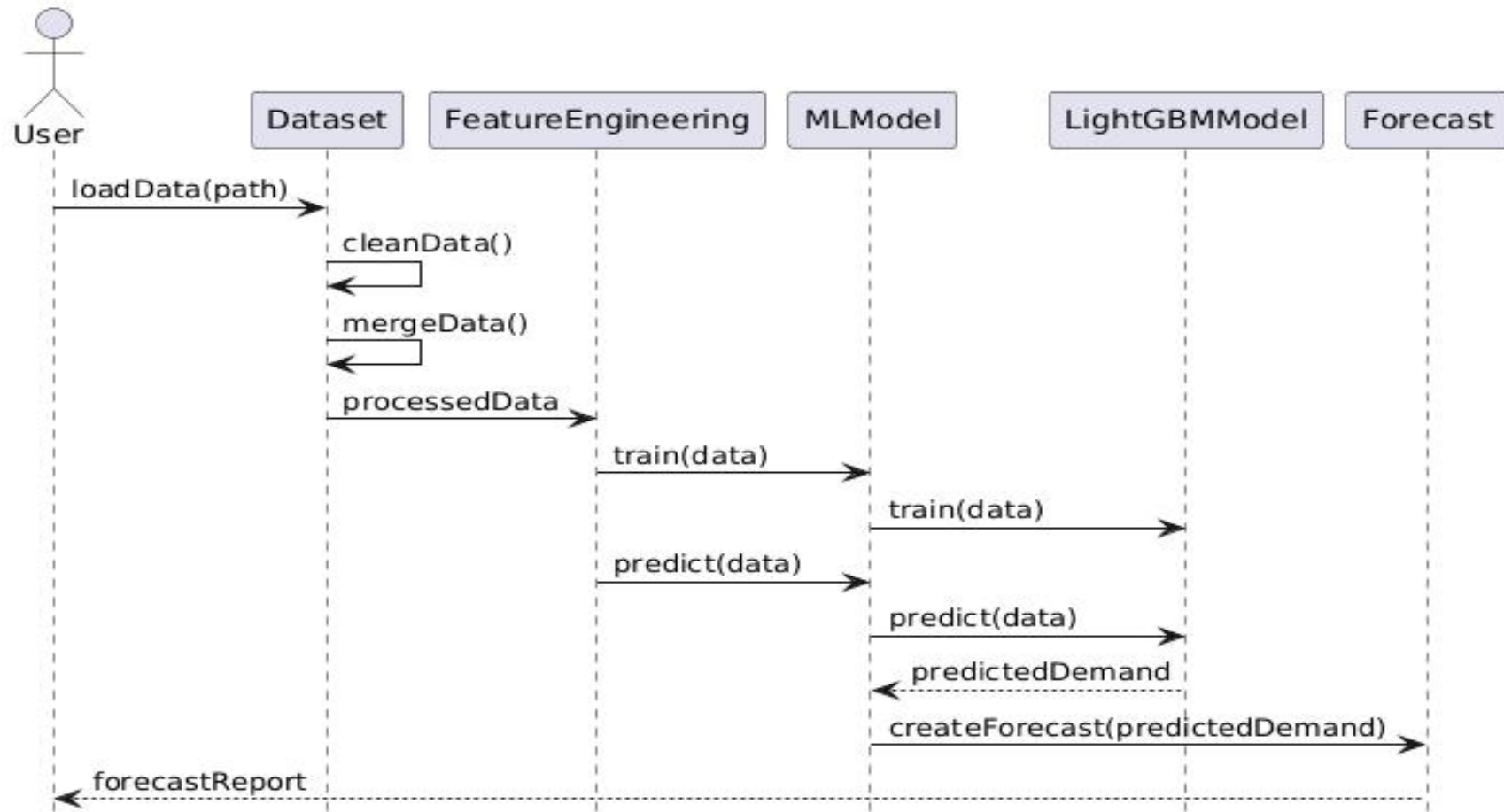
Design

Use Case Diagram



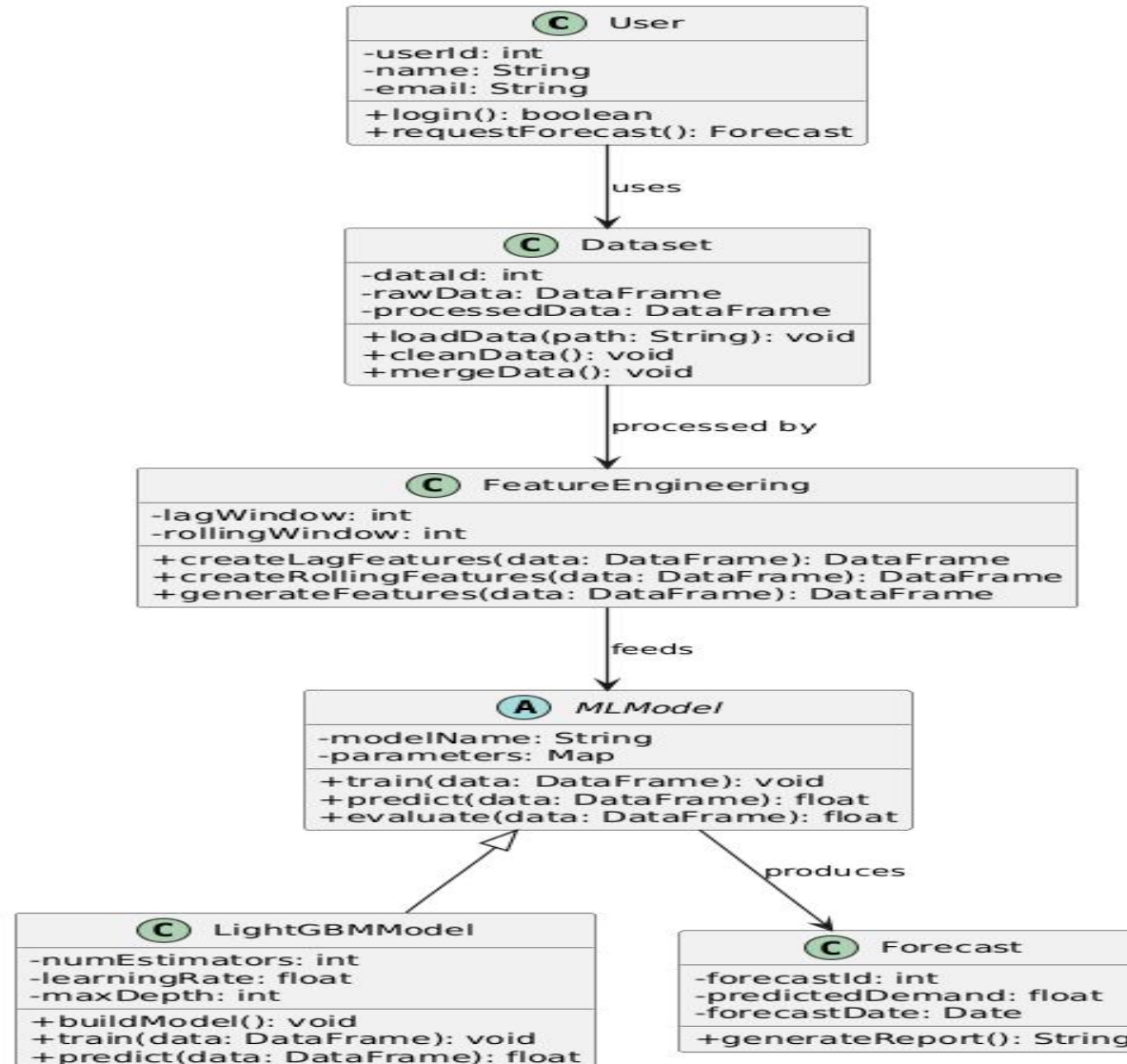
Design

Sequence Diagram



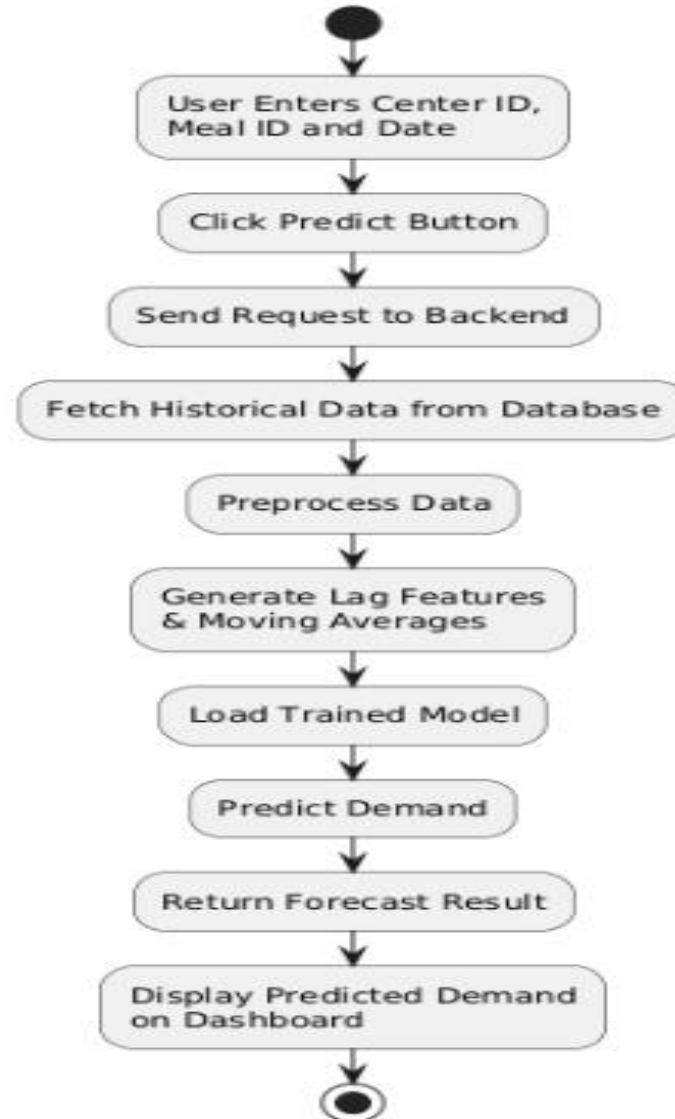
Design

Class Diagram



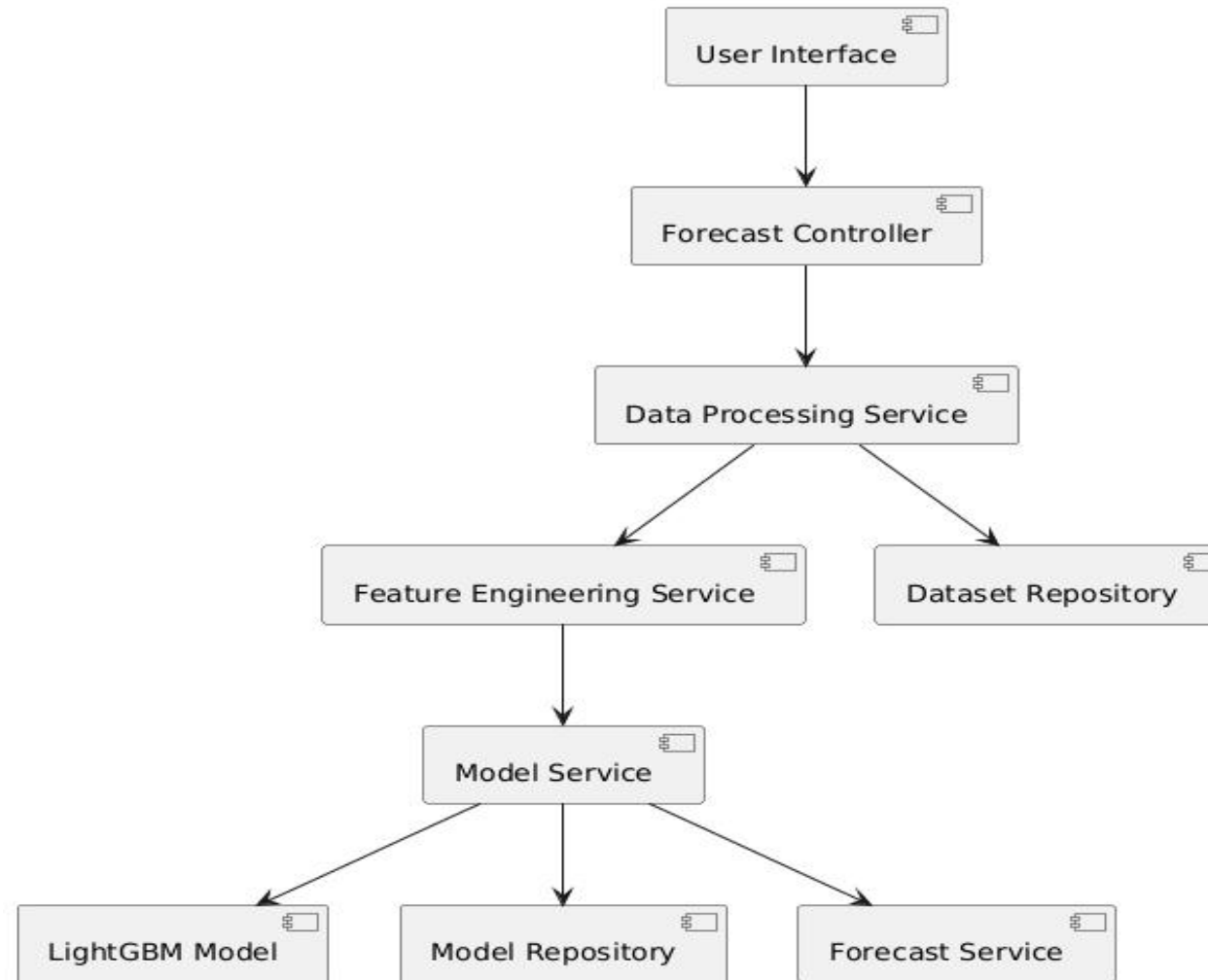
Design

Activity Diagram



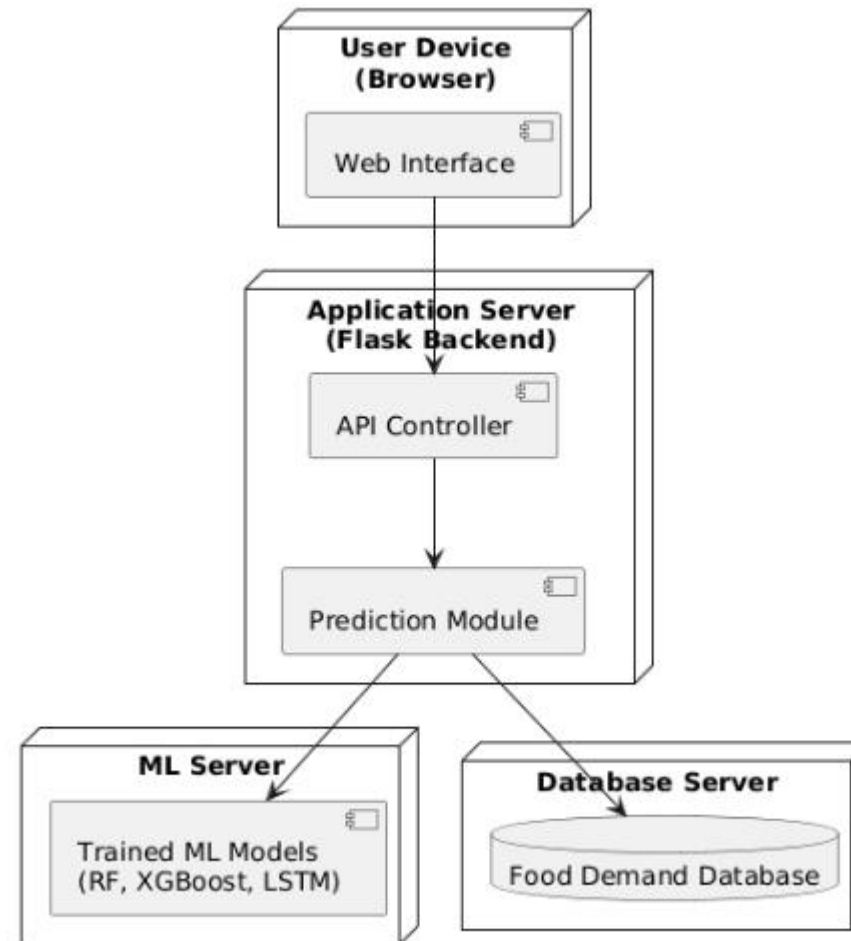
Design

Component Diagram



Design

Deployment Diagram



Implementation/Coding

- Developed using Python, Pandas, NumPy, and Machine Learning models
- **Models implemented:** Random Forest, CatBoost, LightGBM (best performing)
- **Data processed through:**
 - Cleaning & merging datasets
 - Feature engineering (lag features, rolling mean, EWMA)
- **Backend:** Built using Flask API to handle user requests and return predictions
- **Frontend:** Designed using HTML, CSS, JavaScript for user interaction
- **User Inputs:**
 - Center ID, Meal ID, Week, Model selection
- **System Workflow:**

User Input → Feature Generation → Model Prediction → Result Display
- Provides real-time demand prediction, trend analysis, and visualization

Testing Procedures

- Unit testing of individual components (data loading, feature creation)
- API testing using sample JSON inputs
- Frontend testing for user input validation
- Model validation using test dataset

Test Cases

Test Case Number	Test Case	Input	Expected Output	Status
1	Normal Prediction	center_id=11, meal_id=1885, week=146	Prediction in realistic range	Passed
2	Model Comparision	Same input with RF, CatBoost, LightGBM	Same input with RF, CatBoost, LightGBM	Passed
3	Multi-week Forecast	Forecast = 2 or 4 weeks or 8 weeks	Sequential predictions without sudden spikes	Passed
4	Meal Selection	Select center → choose meal	Only valid meals displayed for that center	Passed

- ☒ Backend online — predictions from your trained model

Forecast Parameters

FULFILLMENT CENTER

Center 50 — TYPE_A, City 556

MEAL

1770 — Biryani (Indian)

44 meals available at this center

FORECAST HORIZON

1 Week Ahead — Week 147

Based on Week 146 (last predicted week)

ML MODEL

Random Forest

CatBoost

LightGBM

 Predict Demand

Prediction Result

C50 · BIRYANI (INDIAN) · +1W

LightGBM

PREDICTED FOOD ORDERS

29

units for the selected week

Forecasting Week 146 (base: Week 145)

Confidence range: 28 – 30 units

MODEL ACCURACY

91.2%

on validation set

TREND

-27.5%

vs last known week

FORECAST

+1

weeks ahead

HISTORICAL DEMAND → PREDICTION



● Historical (actual from dataset) ● Model prediction

Results and Inferences

- Random Forest: $R^2 \approx 0.82$
- CatBoost: $R^2 \approx 0.88$
- LightGBM: $R^2 \approx 0.89$ (Best Performance)
- Predictions fall within realistic range (100–2000 units)
- System provides stable results with small confidence range
- LightGBM performs best due to better handling of complex patterns
- Feature engineering (lag, rolling mean) improves accuracy significantly
- Demand is influenced by:
 - Pricing & discounts
 - Promotions
 - Historical demand trends
- Enables accurate forecasting and better inventory planning

Conclusion and Usability

Conclusion :

The Smart Food Demand Forecasting system effectively predicts future food demand using machine learning techniques. Among the models used, LightGBM provides the most accurate and stable results. The use of feature engineering techniques such as lag features and moving averages significantly improves prediction performance. Overall, the system helps reduce food wastage and supports better inventory and supply chain decision-making.

Usability :

- Helps restaurants and food delivery services predict demand accurately
- Supports better inventory planning and reduces food wastage
- Enables data-driven decision-making for supply chain management
- Improves operational efficiency and customer satisfaction

References

- [1] Kaggle, “Genpact – Food Demand Forecasting Competition.” [Online]. Available: <https://www.kaggle.com/competitions/food-demand-forecasting> Accessed: Mar. 1, 2026.
- [2] Microsoft, “LightGBM Documentation.” [Online]. Available: <https://lightgbm.readthedocs.io/>. Accessed: Mar. 1, 2026.
- [3] L. Prokhorenkova, G. Gusev, A. Vorobev, A. V. Dorogush, and A. Gulin, “CatBoost: Unbiased Boosting with Categorical Features,” *Advances in Neural Information Processing Systems*, vol. 31, 2018. [Online]. Available: <https://arxiv.org/abs/1810.11363>. Accessed: Mar. 1, 2026.
- [4] Blue Yonder, “Demand Planning.” [Online]. Available: <https://blueyonder.com/solutions/supply-chain-planning/demand-planning>. Accessed: Mar. 1, 2026.
- [5] Oracle Corporation, “Oracle Demantra Demand Management Overview.” [Online]. Available: https://docs.oracle.com/cd/E18727_01/doc.121/e05179/T493571T493575.htm. Accessed: Mar. 1, 2026.

Future Course of Action

- Improve model accuracy by adding more historical and external data.
- Deploy the system as a cloud-based web application.
- Add real-time demand prediction features.
- Develop a better dashboard for visualization and reports.
- Expand the system to support multiple cities and fulfillment centers.

THANK YOU